

**Owned Requirements:** FR-019, FR-021–FR-025

**Owned Components:** Notification Module, Feedback Module, Admin/Reporting Module

## 3.4 Component Descriptions

### 3.4.1 Notification Module

The Notification Module creates an in-app notification for the affected users whenever an important complaint lifecycle event occurs: submission, assignment, status change, resolution and closure. Users can list their own notifications and mark them as read. A failed notification is logged and never blocks the complaint action that triggered it. The module implements FR-019.

### 3.4.2 Feedback Module

The Feedback Module accepts a rating and an optional comment from the resident who owns a complaint once it reaches Resolved status. It validates eligibility and stores the feedback linked to that complaint, so it can be viewed with the complaint and summarised in reports. The module implements FR-021 and FR-022.

### 3.4.3 Admin/Reporting Module

The Admin/Reporting Module gives the Administrator account management for residents and staff (create staff, change role or department, activate or deactivate), a dashboard of complaint counts by status, and basic reports. Accounts are deactivated rather than deleted so complaint history is preserved, and the module is read-only on complaint data. It implements FR-023 to FR-025.

## 3.7 Risks & Mitigations

| Risk                                                        | Mitigation                                                                       |
|-------------------------------------------------------------|----------------------------------------------------------------------------------|
| A notification fails, so a resident misses a status change. | Failures are logged; the complaint page always shows current status and history. |

## 3.8 Traceability to Requirements

The following rows cover the requirements owned by Rohan and map each requirement to its component and test cases.

| Requirement ID | Component              | Test Case IDs        |
|----------------|------------------------|----------------------|
| FR-019         | Notification Module    | TC-NTF-01, TC-NTF-02 |
| FR-021         | Feedback Module        | TC-FBK-01, TC-FBK-02 |
| FR-022         | Feedback Module        | TC-FBK-03, TC-FBK-04 |
| FR-023         | Admin/Reporting Module | TC-ADM-01, TC-ADM-02 |
| FR-024         | Admin/Reporting Module | TC-ADM-03, TC-ADM-04 |
| FR-025         | Admin/Reporting Module | TC-ADM-05, TC-ADM-06 |

## 4.3 API Design

The following are proposed design-level API interfaces for the Notification, Feedback and Admin/Reporting components. All endpoints require an authenticated session.

### Notification Module

| Endpoint                                  | Method | Purpose                           |
|-------------------------------------------|--------|-----------------------------------|
| /api/notifications                        | GET    | List the user's own notifications |
| /api/notifications/{notification_id}/read | PATCH  | Mark a notification as read       |

### Internal interface: notify()

**Request:** event, complaint\_id, recipient\_ids. Called by the Complaint, Category/Assignment and Processing/Status Modules after their change is saved.

**Response:** One notification record per recipient. The resident is notified of every event; assigned staff are notified on assignment.

**Errors:** Failure is logged; the calling action is not rolled back.

### GET /api/notifications

**Request:** Authenticated session; optional unread=true filter.

**Response:** List of notifications containing notification\_id, complaint\_id, message, is\_read and created\_at, newest first.

**Errors:** 401 – invalid or expired session.

### PATCH /api/notifications/{notification\_id}/read

**Request:** notification\_id path parameter and authenticated session.

**Response:** Notification marked as read.

**Errors:** 401 – user not authenticated; 403 – notification belongs to another user; 404 – notification not found.

## Feedback Module

| Endpoint                                | Method | Purpose                                  |
|-----------------------------------------|--------|------------------------------------------|
| /api/complaints/{complaint_id}/feedback | POST   | Submit feedback for a resolved complaint |
| /api/complaints/{complaint_id}/feedback | GET    | Retrieve feedback for a complaint        |

### POST /api/complaints/{complaint\_id}/feedback

**Request:** rating (1–5), comment (optional)

**Response:** Feedback-recorded message with feedback\_id, complaint\_id, rating and comment.

**Errors:** 400 – rating missing or outside 1–5; 401 – user not authenticated; 403 – user is not the complaint owner; 404 – complaint not found; 409 – complaint not yet Resolved or feedback already submitted.

### GET /api/complaints/{complaint\_id}/feedback

**Request:** complaint\_id path parameter and authenticated session.

**Response:** Feedback details containing feedback\_id, complaint\_id, rating, comment and created\_at.

**Errors:** 401 – user not authenticated; 403 – user not permitted to access the complaint; 404 – no feedback for this complaint.

## Admin/Reporting Module

| Endpoint         | Method | Purpose                      |
|------------------|--------|------------------------------|
| /api/admin/users | GET    | List resident/staff accounts |

| Endpoint                   | Method | Purpose                                  |
|----------------------------|--------|------------------------------------------|
| /api/admin/users           | POST   | Create a staff or administrator account  |
| /api/admin/users/{user_id} | PATCH  | Update role, department or active status |
| /api/admin/dashboard       | GET    | Complaint counts by status               |
| /api/admin/reports         | GET    | Basic reports/statistics                 |

### GET /api/admin/users

**Request:** Optional role and active filters.

**Response:** List of users containing user\_id, name, email, role, department and is\_active. Passwords are never returned.

**Errors:** 401 – user not authenticated; 403 – not an Administrator.

### POST /api/admin/users

**Request:** name, email, password, contact\_number, role, department\_id (staff only)

**Response:** Account-created message and generated user\_id.

**Errors:** 400 – missing/invalid required information; 403 – not an Administrator; 409 – email already registered.

### PATCH /api/admin/users/{user\_id}

**Request:** Any of role, department\_id, is\_active.

**Response:** Updated user record.

**Errors:** 400 – invalid value; 403 – not an Administrator; 404 – user not found; 409 – administrator cannot deactivate their own account.

### GET /api/admin/dashboard

**Request:** Authenticated Administrator session.

**Response:** Total complaints, count per status (Submitted, Assigned, In Progress, Resolved, Closed), pending count and average rating.

**Errors:** 401 – user not authenticated; 403 – not an Administrator.

### GET /api/admin/reports

**Request:** type (status, category, period, pending or workload); optional from and to dates.

**Response:** Report type and rows of label and count. For workload, the label is the staff member or department.

**Errors:** 400 – unknown type or from later than to; 403 – not an Administrator.

## 4.5 UX Design

| Screen          | User          | Design                                                                        |
|-----------------|---------------|-------------------------------------------------------------------------------|
| Notifications   | All roles     | Bell icon with unread count; list newest first, linked to the complaint.      |
| Feedback form   | Resident      | Shown once Resolved: 1–5 stars, optional comment; read-only after submission. |
| Admin dashboard | Administrator | Count cards per status, total, pending and average rating.                    |
| User management | Administrator | User table with role filter; Add Staff, Edit, Activate/Deactivate. No delete. |
| Reports         | Administrator | Report type and date range selectors; results table.                          |